

Semiconductor Sector — Fundamental Analysis

120 companies ranked by irreplaceability in the value chain. The "toll booth" criterion: whose disappearance would halt the industry?

Analysis date: August 6, 2026 · Investment horizon: 40 years · Framework: \$64–\$67

COMPANIES ANALYZED

120

Across full value chain

TIER 1 — IRREPLACEABLE

15

Industry halts without them

TIER 2 — CRITICAL

25

Very hard to substitute

TIER 3 — IMPORTANT

40

Strong but substitutable

TIER 4 — RELEVANT

40

Competitive but valuable

AGGREGATE MKT CAP

~\$8.5T

Listed companies only

Methodology: Irreplaceability Score (0–100)

The ranking criterion is **insustituibilidad en la cadena de valor** — not size, not margins, not revenue. A company scores high when its disappearance would create a bottleneck that no competitor could fill within 3–5 years. The score weights:

- **Monopoly/oligopoly position (35%)** — Market share concentration and number of viable alternatives
- **Technical moat depth (25%)** — Proprietary technology, IP, know-how that would take a decade+ to replicate
- **Supply chain criticality (20%)** — Position in the dependency graph; how many downstream nodes depend on this company
- **Switching costs (10%)** — Customer lock-in through design wins, qualification cycles, ecosystem integration
- **Geopolitical uniqueness (10%)** — Geographic concentration of capability; strategic national importance

Financial health is assessed separately: **Strong** (all key ratios healthy), **Good** (mostly healthy), **Moderate** (some concerns), **Weak** (significant issues).

TIER 1

Irreplaceable — Industry Halts Without Them

These 15 companies occupy positions so dominant and technically deep that their sudden absence would shut down or severely cripple semiconductor manufacturing globally. True toll booths.

#	COMPANY	SEGMENT	MKT CAP	REV (TTM)	GROSS %	OP %	P/E	EV/EBITDA	D/E	ROE %	ROIC %	FCF YIELD	HEALTH	IRREPL.	MOATS	RATIONALE
1	TSMC TSM	Foundry	\$950B	\$95B	56%	45%	27x	17x	0.25	30%	23%	2.8%	STRONG	98	ScaleIPSwitching	Fabricates ~92% of sub-5nm chips. No alternative exists at scale. Apple, NVIDIA, AMD, Qualcomm all depend on TSMC. Would take \$200B+ and 10+ years to replicate.
2	ASML ASML	Lithography Equipment	\$340B	\$32B	51%	36%	32x	26x	0.35	68%	32%	2.2%	STRONG	99	MonopolyIP	Sole maker of EUV lithography systems. 100% market share in EUV. Without ASML, no sub-7nm chips exist.

	#	Company	Segment	Mkt Cap	Rev (TTM)	Gross %	Op %	P/E	EV/EBITDA	D/E	ROE %	ROIC %	FCF Yield	Health	Irrepl.	Moats	Rationale
																	The ultimate toll booth in semiconductors.
	3	NVIDIA NVDA	AI/GPU Design	\$3,200B	\$130B	75%	62%	55x	45x	0.40	120%	45%	1.5%	STRONG	96	EcosystemIPNetwork	CUDA ecosystem lock-in makes GPU+software stack irreplaceable for AI training. ~90% share of AI training accelerators. Valuation stretched but position unassailable near-term.
	4	Synopsys SNPS	EDA Software	\$85B	\$7.0B	80%	30%	55x	40x	0.20	25%	18%	1.6%	STRONG	95	DuopolySwitchingIP	Duopoly with Cadence in EDA. Every chip designed on Earth uses Synopsys or Cadence tools. Qualification cycles of 5-10 years. No chip exists without EDA.
	5	Cadence Design CDNS	EDA Software	\$80B	\$4.8B	89%	32%	65x	50x	0.30	32%	22%	1.3%	STRONG	94	DuopolySwitchingIP	Other half of the EDA duopoly. Strongest in analog/mixed-signal and verification. Same irreplaceability logic as Synopsys — no chips without EDA tools.
	6	Arm Holdings ARM	IP/Architecture	\$160B	\$3.8B	96%	25%	100x+	85x	0.15	18%	14%	0.8%	GOOD	95	MonopolyNetworkIP	99%+ of mobile processors use Arm architecture. Expanding into data center, automotive. The instruction set is embedded in billions of devices — switching is near-impossible. Valuation extreme.
	7	Lam Research LRCX	Etch/Deposition Equipment	\$110B	\$17B	47%	31%	25x	20x	0.55	55%	28%	2.8%	STRONG	93	OligopolyIPSwitching	Dominant in etch and deposition — the most process step-intensive part of chipmaking. Critical for 3D NAND (100+ layers). Oligopoly with TEL/AMAT.

#	COMPANY	SEGMENT	MKT CAP	REV	GROSS %	OP %	P/E	D/E	ROIC %	FCF Y.	HEALTH	IRREPL.	RATIONALE
31	Wolfspeed WOLF	SiC Substrates	\$3B	\$0.8B	15%	-35%	N/A	3.50	-15%	-12%	WEAK	83	Largest SiC substrate maker globally. SiC is critical for EV power electronics and grid infrastructure. Financial distress (high debt, losses) but the technology is irreplaceable. High risk/high reward.
32	Photronics PLAB	Photomasks	\$2B	\$1.0B	35%	22%	10x	0.05	18%	8.0%	STRONG	80	One of only 3 companies that make advanced photomasks (with Toppan and DNP). Every chip design requires a mask set. \$2B market cap for a critical infrastructure player. Cheap.
33	Kokusai Electric 6525.T	Batch Deposition Equipment	\$5B	\$2B	45%	22%	18x	0.20	20%	4.0%	GOOD	79	Dominant in batch ALD/CVD deposition equipment for 3D NAND. As NAND layers increase (200+), Kokusai's batch tools become more critical. Re-IPO'd after Applied Materials deal blocked.
34	MediaTek 2454.TW	Mobile/IoT SoC	\$65B	\$18B	48%	18%	16x	0.10	22%	4.5%	STRONG	76	#1 smartphone SoC by unit volume globally. Powers ~40% of all smartphones. Critical for mid-range/emerging market connectivity. Not as "irreplaceable" as Qualcomm's IP but massive reach.
35	AMD AMD	CPU/GPU/AI Accel.	\$230B	\$28B	50%	22%	28x	0.05	8%	2.8%	STRONG	78	Only credible x86 competitor to Intel. EPYC dominates server market share gains. MI300 AI accelerators provide the only alternative to NVIDIA at scale. Competitive necessity for the industry.
36	Intel INTC	CPU/Foundry (IDM)	\$90B	\$52B	40%	-5%	N/A	0.50	-2%	-1%	WEAK	80	Geopolitically essential — only Western company with leading-edge foundry ambitions. \$20B+ CHIPS Act subsidies. Financial stress (losses, margin compression) but strategic irreplaceability for US/EU supply chain independence.
37	Advantest 6857.T	Test Equipment	\$35B	\$5B	58%	30%	35x	0.05	28%	2.5%	STRONG	83	~50% of semiconductor test equipment (duopoly with Teradyne). Every chip must be tested. AI chips = more complex testing = more Advantest content per wafer. Zero debt.
38	Teradyne TER	Test Equipment	\$20B	\$3B	58%	25%	28x	0.10	22%	3.0%	STRONG	80	Other half of the test duopoly. Strong in analog/mixed-signal testing and industrial robotics (Universal Robots). Every chip must pass test — no shortcuts.
39	ON Semiconductor ON	Power/SiC/Image Sensors	\$30B	\$7B	45%	28%	15x	0.40	20%	5.5%	STRONG	76	#2 in SiC (behind Wolfspeed), #2 in auto image sensors (behind Sony). EV power modules with 10+ year LTSAs. Transformed from commodity to specialized player. Strong execution.
40	Monolithic Power Systems MPWR	Power Management ICs	\$40B	\$2.5B	56%	28%	45x	0.00	30%	1.8%	STRONG	76	Fastest-growing power IC company. Critical for AI server power delivery. NVIDIA GPU boards use MPWR voltage regulators. Zero debt, 30% ROIC. Premium but earned.

TIER 3

Important — Strong Position, Substitutable Within 1–3 Years

Market leaders or strong #2/#3 players in their segments. Valuable franchises with real moats but alternatives exist or could emerge. Many are excellent long-term investments despite lower irreplaceability.

[illegible]

#	COMPANY	SEGMENT	MKT CAP	REV	P/E	D/E	ROIC	FCF Y.	HEALTH	IRREPL.	RATIONALE
55	Skyworks Solutions SWKS	RF Front-End	\$15B	\$4.5B	14x	0.10	20%	5.5%	STRONG	67	Major RF front-end supplier for smartphones (Apple ~55% of revenue). BAW filters, power amplifiers. Apple concentration is a risk but also a moat (qualified supplier).
56	Qorvo QRVO	RF/GaN/BAW	\$7B	\$3.5B	15x	0.50	10%	6.0%	GOOD	66	GaN expertise for defense/infrastructure, BAW filters for 5G. Diversifying from mobile to defense and AI infrastructure power. Cheap but turnaround uncertain.
57	Mobileye MBLY	ADAS/Autonomous Vision	\$12B	\$2.0B	50x	0.00	8%	1.5%	GOOD	73	~70% ADAS market share by volume. EyeQ chips in 150M+ vehicles. Design wins extend 5+ years. Autonomous driving = more silicon content per car. Intel subsidiary.
58	SCREEN Holdings 7735.T	Wet Clean Equipment	\$8B	\$4B	12x	0.15	18%	5.0%	STRONG	75	~60% share of single-wafer wet cleaning tools. Every fab uses SCREEN cleaners after nearly every process step. Hidden champion at 12x P/E. Cleaning steps increase with node shrinks.
59	Edwards Vacuum ATLKY (parent)	Vacuum/Abatement	\$45B*	~\$3B*	~25x	0.30	~15%	~3%	GOOD	74	~40% of fab vacuum and gas abatement systems. Every deposition/etch/implant step requires vacuum. Atlas Copco subsidiary — invest via parent. *Segment estimated.
60	VAT Group VACN.SW	Vacuum Valves	\$12B	\$1.0B	40x	0.20	25%	2.0%	STRONG	76	~70% global share of high-vacuum valves for semiconductor tools. Every AMAT, Lam, TEL tool needs VAT valves. Swiss precision, near-monopoly in niche. Premium valuation deserved.
61	Hamamatsu Photonics 6965.T	Photosensors/Detectors	\$5B	\$1.5B	20x	0.05	12%	3.5%	STRONG	71	Dominant in photomultiplier tubes and specialized photosensors. Critical for KLA inspection systems and medical imaging. Niche monopoly positions across photonics.
62	Brooks Automation AZTA	Wafer Handling/Automation	\$4B	\$0.7B	35x	0.10	10%	3.0%	GOOD	68	Wafer-handling robots and contamination control. Renamed Azena. Fabs are cleanrooms — automated wafer transport is non-negotiable. Niche but essential.
63	Onto Innovation ONTO	Inspection/Metrology	\$8B	\$1.0B	30x	0.00	18%	2.5%	STRONG	69	Advanced packaging inspection and 3D metrology. Complements KLA in emerging packaging inspection. Zero debt. Positioned for chiplet/heterogeneous integration trend.
64	Cohu COHU	Test Handlers/Contactors	\$1.5B	\$0.5B	20x	0.15	12%	5.0%	GOOD	65	Test handlers and contactors that complement Advantest/Teradyne testers. Niche but necessary — you can't test a chip without handling it. Automotive grade testing strength.
65	Hirose Electric 6806.T	Connectors	\$6B	\$2B	18x	0.00	15%	4.0%	STRONG	65	Premium board-to-board connectors for test equipment and mobile devices. Apple supplier. Japanese precision. Zero debt. Niche connector monopoly positions.
66	Kulicke & Soffa KLIC	Wire Bonding/Packaging Equipment	\$3B	\$0.7B	15x	0.00	15%	6.0%	STRONG	68	~60% of wire bonding equipment globally. Wire bonding still used in ~75% of all semiconductor packages. Transitioning to advanced packaging (thermoccompression). Cheap, zero debt.
67	PDF Solutions PDFS	Yield Analytics Software	\$1.5B	\$0.2B	60x	0.00	10%	2.0%	GOOD	67	AI-driven yield improvement software used by leading fabs. As nodes shrink, yield analytics becomes more critical. Small but growing recurring revenue model. Niche SaaS for fabs.

#	Company	Segment	Mkt Cap	Rev	P/E	D/E	ROIC	FCF Y.	Health	Irrepl.	Rationale
68	Nova Ltd NVMI	Metrology	\$6B	\$0.7B	30x	0.00	22%	3.0%	STRONG	69	Optical and X-ray metrology for advanced nodes. In-line CD measurement. Israeli precision engineering. Complementary to KLA. Zero debt, strong margins.
69	Camtek CAMT	Inspection (Packaging)	\$4B	\$0.4B	30x	0.00	25%	3.0%	STRONG	67	Automated optical inspection for advanced packaging, bumps, and HBM. Only pure-play packaging inspection company. Beneficiary of chiplet revolution. Zero debt.
70	FormFactor FORM	Probe Cards	\$3B	\$0.7B	25x	0.05	15%	4.0%	GOOD	70	~50% of advanced probe cards (wafer test interface). Every chip must be probed before packaging. MEMS-based probe cards for advanced nodes. Consumable with recurring revenue.
71	Murata Manufacturing 6981.T	MLCCs/Passives	\$30B	\$14B	15x	0.05	12%	5.0%	STRONG	72	World's #1 MLCC (ceramic capacitor) maker. Every electronic device needs dozens-to-thousands of MLCCs. An iPhone uses 1,000+. Murata's precision at tiny form factors is moat-worthy.
72	TDK Corporation 6762.T	Passives/Sensors/Batteries	\$20B	\$14B	14x	0.20	10%	4.0%	GOOD	66	Ferrite cores, inductors, MEMS sensors, micro-batteries. Broad passive component portfolio. ADAS sensor suite strength. Diversified Japanese electronics stalwart.
73	Siltronic WAF.DE	Silicon Wafers	\$4B	\$1.5B	10x	0.10	12%	6.0%	GOOD	72	#4 wafer maker globally (~15% share). European strategic wafer supplier. GlobalWafers bid failed — remains independent. Cheap at 10x. Cyclical but structurally important.
74	Soitec SOI.PA	SOI/Advanced Substrates	\$4B	\$1.0B	20x	0.30	12%	3.5%	GOOD	73	Near-monopoly in SOI (silicon-on-insulator) wafers. FD-SOI used for low-power IoT, 5G RF, automotive. Also GaN-on-Si substrates. French champion in advanced materials.
75	Allegro MicroSystems ALGM	Magnetic Sensors/Power	\$5B	\$1.0B	25x	0.10	15%	4.0%	GOOD	65	Market leader in magnetic current sensors for EV powertrains. Every EV motor controller needs current sensing. Secular EV growth driver. Clean balance sheet.
76	Vishay Intertechnology VSH	Passives/Discretes	\$3B	\$3.5B	8x	0.25	8%	6.0%	GOOD	62	Broadest passive/discrete portfolio. Resistors, MOSFETs, diodes used everywhere. Not dominant in any one area but breadth is the moat. Very cheap at 8x P/E. Value play.
77	Sensata Technologies ST	Sensors/Industrial	\$5B	\$4B	12x	1.20	10%	6.0%	MODERATE	63	Mission-critical sensors (pressure, temp, position) for auto and industrial. 800+ SKUs embedded in vehicles. D/E a concern but FCF strong. Electrification = more sensors per vehicle.
78	Navitas Semiconductor NVT.S	GaN Power ICs	\$1.5B	\$0.1B	N/A	0.00	-20%	-15%	WEAK	66	GaN power IC pioneer — integrates GaN power + driver + control. GaN is the future of fast chargers, data center power, EV on-board chargers. Pre-profit but technology moat real.
79	Air Liquide AI.PA	Process Gases	\$100B	\$28B	28x	0.45	12%	3.0%	STRONG	73	Duopoly with Linde in process gases. On-site gas generation for TSMC, Samsung, Intel fabs. 15-20 year contracts. ~10% of revenue from electronics. Investable through parent.
80	JSR Corporation Acquired by JIC	Photoresists	Private	~\$3B	—	—	—	—	N/A	80	Major photoresist and CMP slurry supplier. Taken private by Japan Investment Corp — no longer directly investable. Included for value chain context. Invest via peers (TOK, Fujifilm).

#	COMPANY	SEGMENT	MKT CAP	P/E	D/E	HEALTH	IRREPL.	RATIONALE
95	Veeco Instruments VECO	Deposition (MBE/Laser Anneal)	\$2B	15x	0.30	GOOD	62	Laser spike anneal and ion beam deposition for advanced nodes. Niche process tools. Growing EUV pellicle deposition capability. Small but positioned at critical process steps.
96	MKS Instruments MKSI	Instruments/Subsystems	\$8B	20x	1.50	MODERATE	65	Power supplies, pressure gauges, gas flow controllers — the plumbing of every fab tool. D/E elevated from Atotech acquisition. Technology is essential but substitutable. Deleveraging.
97	Aixtron AIXA.DE	MOCVD Equipment	\$3B	15x	0.00	STRONG	65	Leading MOCVD equipment for GaN and SiC epitaxy. Key enabler for power semiconductors and micro-LEDs. Duopoly with Veeco in some segments. Zero debt. European equipment champion.
98	Nordson ASYMTEK NDSN	Dispensing/Bonding	\$14B	25x	0.50	STRONG	58	Precision dispensing and bonding equipment for semiconductor packaging. Also test/inspection. Part of diversified Nordson — semi exposure ~30%. Steady compounder.
99	Photon Dynamics	Acquired by Orbotech → KLA. Access via KLA (Tier 1 #9).						
100	Ultra Clean Holdings UCTT	Gas Delivery/Cleaning	\$2B	18x	0.40	GOOD	58	Gas delivery systems and precision cleaning for fabs. Critical contamination control. Customer-qualified = switching costs. Small-cap semi infrastructure play.
101	Photronics	Already in Tier 2 #32 — listed for photomask criticality.						
102	Ichor Holdings ICHR	Fluid Delivery	\$1B	20x	0.30	GOOD	55	Gas and chemical delivery subsystems for Lam, AMAT, TEL. They build the "plumbing" that OEMs integrate. Supplier-to-the-suppliers model. Cyclical but essential infrastructure.
103	Indie Semiconductor INDI	Auto ADAS/Sensors	\$1.5B	N/A	0.40	MODERATE	55	ADAS radar, LiDAR, and vision processors. Pre-profit but strong design wins with major OEMs. Pure-play ADAS at lower valuations than Mobileye. Growth story.
104	Wolfspeed	See Tier 2 #31 — technology-critical despite financial weakness.						
105	MaxLinear MXL	Broadband/Infrastructure ICs	\$2B	25x	0.50	MODERATE	52	Broadband access (PAM4 DSP), optical interconnect. Data center and telecom infrastructure. Niche positions. Volatile but positioned for bandwidth growth.
106	Diodes Incorporated DIOD	Discrete/Analog	\$3B	15x	0.20	GOOD	50	Standard discrete and analog components. Commodity-adjacent but essential for consumer, auto, industrial. Value play — fundamental building blocks.
107	Tower Semiconductor TSEM	Specialty Foundry	\$6B	16x	0.10	GOOD	60	Specialty foundry (RF-SOI, SiGe, BCD power). Israeli fab operations. Intel partnership for 300mm expansion. Not a leading-edge foundry but analog/RF specialty is sticky.
108	Vnet Group / GDS	Data Center Infra (China)	~\$5B	N/A	High	WEAK	45	Chinese data center operators. Adjacent to semiconductors via demand for server chips. Geopolitical risk, high leverage. Included for China exposure completeness.
109	SMIC 0981.HK	Foundry (China)	\$40B	20x	0.15	MODERATE	58	China's most advanced foundry (~7nm without EUV). Sanctioned — can't get ASML EUV. Geopolitically critical for China's semiconductor independence. Investability limited.
110	Hua Hong Semiconductor 1347.HK	Foundry (China, Mature)	\$5B	15x	0.25	GOOD	48	Chinese specialty foundry (power, NOR flash, BCD). Government-backed. Sanctions risk but mature-node focused = less geopolitical exposure vs SMIC.

#	COMPANY	SEGMENT	MKT CAP	P/E	D/E	HEALTH	IRREPL.	RATIONALE
111	Will Semiconductor 603501.SS	Image Sensors (China)	\$10B	20x	0.15	GOOD	55	OmniVision parent — #3 in image sensors after Sony and Samsung. Strong in mid-range smartphone, auto, and security cameras. Chinese A-share access only.
112	GigaDevice 603986.SS	NOR Flash/MCU (China)	\$5B	30x	0.05	GOOD	48	Chinese NOR flash and ARM MCU leader. Beneficiary of domestic substitution. Growing but competes with established Western/Japanese players. China localization tailwind.
113–120	Also tracked: Winbond (NOR/DRAM, 2344.TW), Nanya Tech (DRAM, 2408.TW), ITEQ (CCL substrates, 6213.TW), Ibiden (IC substrates, 4062.T), Shinko Electric (packaging substrates, 6967.T), Heraeus (bonding wire, private), Fujifilm (photoresists/CMP, 4901.T), Tokyo Ohka Kogyo (photoresists, 4186.T). All serve necessary functions in the value chain; their irreplaceability scores range 45–65 depending on market share concentration and alternatives.							

Key Investment Takeaways

Applying the irreplaceability lens to investment decisions, filtered through the 40-year framework (\$64–\$67). These are not trading ideas — they are structural positions to hold through multiple cycles.

The Untouchable Trio: ASML + TSMC + EDA

ASML, TSMC, Synopsys, and Cadence form the hardest-to-replicate layer of the semiconductor stack. ASML's EUV monopoly is the strongest toll booth in any industry. TSMC's 92% share of sub-5nm is unassailable for a decade. The EDA duopoly hasn't had a credible challenger in 30 years. These are core positions (\$16: 3x evidence to rotate out). Valuation is secondary to irreplaceability over 40 years.

Hidden Champions: Equipment & Materials

Lasertec (EUV mask inspection monopoly), Disco (80% dicing saws), SCREEN (60% wet cleaners), VAT Group (70% vacuum valves), Entegris (purity filtration). These companies are invisible to the public but utterly essential. Most are Japanese/European, trade at reasonable valuations (12–20x P/E), and benefit from every dollar of semiconductor capex regardless of which chip architecture wins.

The Wafer Duopoly: Shin-Etsu + SUMCO

Every chip starts as a silicon wafer. Two companies supply ~55% of the world's 300mm wafers. Shin-Etsu at 15x P/E with near-zero debt is arguably the cheapest "irreplaceable" asset in semiconductors. Qualification cycles of 2+ years make switching virtually impossible. Cyclical but structurally essential.

Best Value-for-Irreplaceability

Companies whose valuation doesn't reflect their criticality: **Shin-Etsu** (15x), **STMicro** (8x), **UMC** (11x), **Photronics** (10x), **SCREEN Holdings** (12x), **K&S** (15x), **Siltronic** (10x), **SK Hynix** (8x). These trade at discounts because they're cyclical, Japanese/European, or unglamorous. The 40-year investor benefits from buying irreplaceability at a discount.

High Irreplaceability, High Risk

Wolfspeed (SiC substrates — critical tech, weak financials), **Intel** (geopolitically essential, operationally struggling), **Entegris** (purity monopoly but D/E from CMC merger). These require conviction and position sizing (satellite, not core — \$15). The technology is irreplaceable but the companies face financial or execution risk.

Spanish Tax Considerations (\$33–\$34)

For fund transferability in Spain, prefer UCITS-domiciled ETFs (SOXX, SMH are US-listed — use iShares Semiconductor ETF UCITS or similar). Individual stocks through a broker with Spanish tax reporting integration. Capital gains on transferred funds (traspaso) are tax-deferred if switching between UCITS funds. Direct stock positions do not benefit from traspaso — consider for core holdings only.

Semiconductor ETFs for Portfolio Construction

For broad semiconductor exposure, ETFs provide instant diversification. The choice between them is a bet on value chain positioning.

SMH — VanEck Semiconductor

Top-heavy (NVDA ~20%, TSM ~12%, AVGO ~8%). Best AI exposure.

SOXX — iShares Semiconductor

More equal-weighted. Broader equipment exposure than SMH.

SOXQ — Invesco PHLX Semiconductor

Cheapest broad semi ETF. Modified equal-weight.

UCITS: IUSG / WTCH

For Spanish tax traspaso — UCITS-domiciled semiconductor funds.

AUM	~\$25B	AUM	~\$15B	AUM	~\$1B	Domicile	Ireland/Luxembourg
Expense Ratio	0.35%	Expense Ratio	0.35%	Expense Ratio	0.19%	Tax	Traspaso eligible
Holdings	26	Holdings	35	Holdings	30	Key Advantage	Fund transferability
5Y Return	~28% CAGR	5Y Return	~25% CAGR	5Y Return	~24% CAGR	Note	Check availability at broker

Suggested Portfolio Architecture

Applying §64–§67 framework. Core positions require 3x evidence to rotate (§16). Satellite allows for higher-conviction bets. All positions evaluated against the 5-year lock-up test (§63).

Core Positions (70% of semi allocation)

Infrastructure monopolies: ASML, TSMC, Synopsys or Cadence, Lam Research, Applied Materials, KLA. **Wafer supply:** Shin-Etsu. **ETF for breadth:** SMH or SOXX (UCITS version for traspaso). These pass the "wouldn't touch for 5 years" test. Valuation entry points matter but structural position matters more over 40 years.

Satellite Positions (30% of semi allocation)

Hidden champions: Lasertec, Disco, SCREEN, VAT Group, BE Semiconductor. **Value plays:** STMicro, UMC, Photonics, SK Hynix. **Thematic:** NVIDIA (AI), Wolfspeed (SiC/EV — position size reflects risk), Credo/Astera (AI interconnect). Satellite allows faster rotation with less evidence (§16).

Disclaimer: This analysis is for educational and research purposes within the long-term investment framework (§64–§67). Financial data represents approximate values from most recent available filings and may not reflect real-time prices. Irreplaceability scores are subjective assessments based on market structure analysis. Past performance does not guarantee future results. Consult a qualified financial advisor before making investment decisions.

Generated: August 6, 2026 · Framework version: §64–§67 · Companies analyzed: 120 · Primary criterion: Insustituibilidad en la cadena de valor